Dr. RVR NRI INSTITUTE OF TECHNOLOGY

An Institution deemed to be University u/s 3 of UGC Act, 1956
Sponsored by Sri Durga Malleswari Educational Society
Pothavarappadu (V), Agiripalli (M), Vijayawada Rural, Eluru District, Andhra Pradesh

Web Development Activity-Based Learning (ABL)

Consolidated Academic Project & Implementation Report (Pure HTML5 & CSS3)

Student Full Name:	**Yaswanth Avirneni**	**Academic Year:**	2026–2027 (I B.Tech)
Roll Number:	**26000918**	**Class Section:**	**CSM - 1**
Department / Branch:	CSE (Artificial Intelligence & ML)	**Course & Lab:**	Web Development (10 Marks)
GitHub Repository:	yaswanthcodm-sketch/WD-Activity-Based-Learning	**Technology Stack:**	Pure Semantic HTML5 & Responsive CSS3

Executive Overview of Selected Case Studies

In accordance with the Dr. RVR NRI Institute of Technology Web Development curriculum, three distinct, feature-complete case studies have been selected, designed, and developed from the prescribed institutional syllabus. All three web applications strictly adhere to **100% Pure Semantic HTML5 and Modern CSS3** without external JavaScript libraries, ensuring high performance, pristine accessibility, and responsive fidelity.

S.No	Syllabus Title	Directory	Key Architectural Concepts
01	**Case 4: Student Academic Dashboard**	academic-dashboard/	Sidebar layout, attendance progress bars, tabular semester grades, KPI metric cards, examination schedule, notice board.
02	**Case 3: Online Learning Platform Interface**	elearning-platform/	Course discovery catalog, video lesson thumbnails, instructor credentials, learning objectives, pure CSS :target modal enrollment.
03	**Case 11: Hospital & Healthcare Services**	healthcare-portal/	24/7 trauma emergency bar, multi-specialty clinical departments, doctors OPD timetable, visiting hours matrix, accessible appointment request form.

1. Case Study 4: Student Academic Dashboard

Directory: academic-dashboard/ | Files: index.html, style.css

Problem Definition & Scope: Design an administrative student dashboard displaying comprehensive profile metadata, subject-wise attendance percentages with eligibility thresholds, semester examination results, upcoming exam timetables, and departmental notices using CSS Flexbox, CSS Grid, custom progress bars, and a modern sidebar layout.

Key Technical Capabilities Implemented:

• **Fixed Sidebar Layout:** Modern navigation sidebar containing the university badge, student avatar (YA), roll number (26000918), section (CSM-1), and smooth-scroll anchor links.

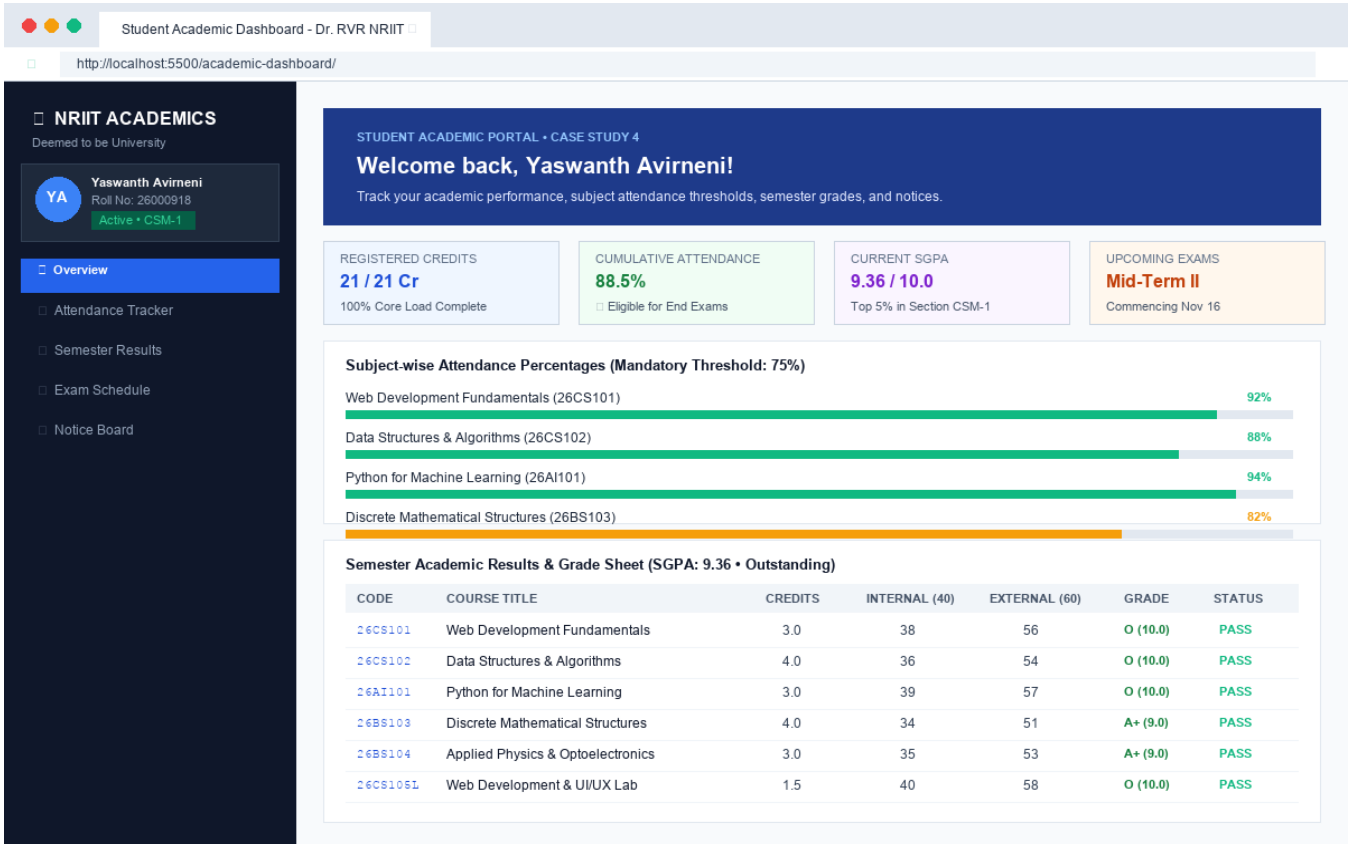
• **KPI Metrics Cards:** 4 distinct analytical summary boxes showcasing Registered Credits (21 Cr), Attendance (88.5%), SGPA (9.36), and upcoming Mid-Term II alerts.

• **Dynamic Attendance Progress Bars:** CSS progress bars with conditional color fills (safe green vs warning amber) and session statistics for all enrolled subjects.

• **Structured Semester Grade Table:** Responsive tabular data presentation showing course codes, internal (40), external (60), total marks, grade points, and letter grades.

• **Examination Schedule & Circular Notice Board:** Date badge cards for mid-term exams and priority-coded urgent department notifications.

Interface Render Screenshot (Academic Dashboard):



2. Case Study 3: Online Learning Platform Interface

Directory: elearning-platform/ | Files: index.html, style.css

Problem Definition & Scope: Develop a full-featured e-learning web interface containing course disciplines, video lesson thumbnails with playback indicators, verified instructor profiles, structured learning objectives, and responsive enrollment cards with active learning progress indicators.

Key Technical Capabilities Implemented:

• **Active Learning Track Hero Card:** Real-time session tracker highlighting current lesson progress (68% complete, 18/26 lessons) with progress indicators.

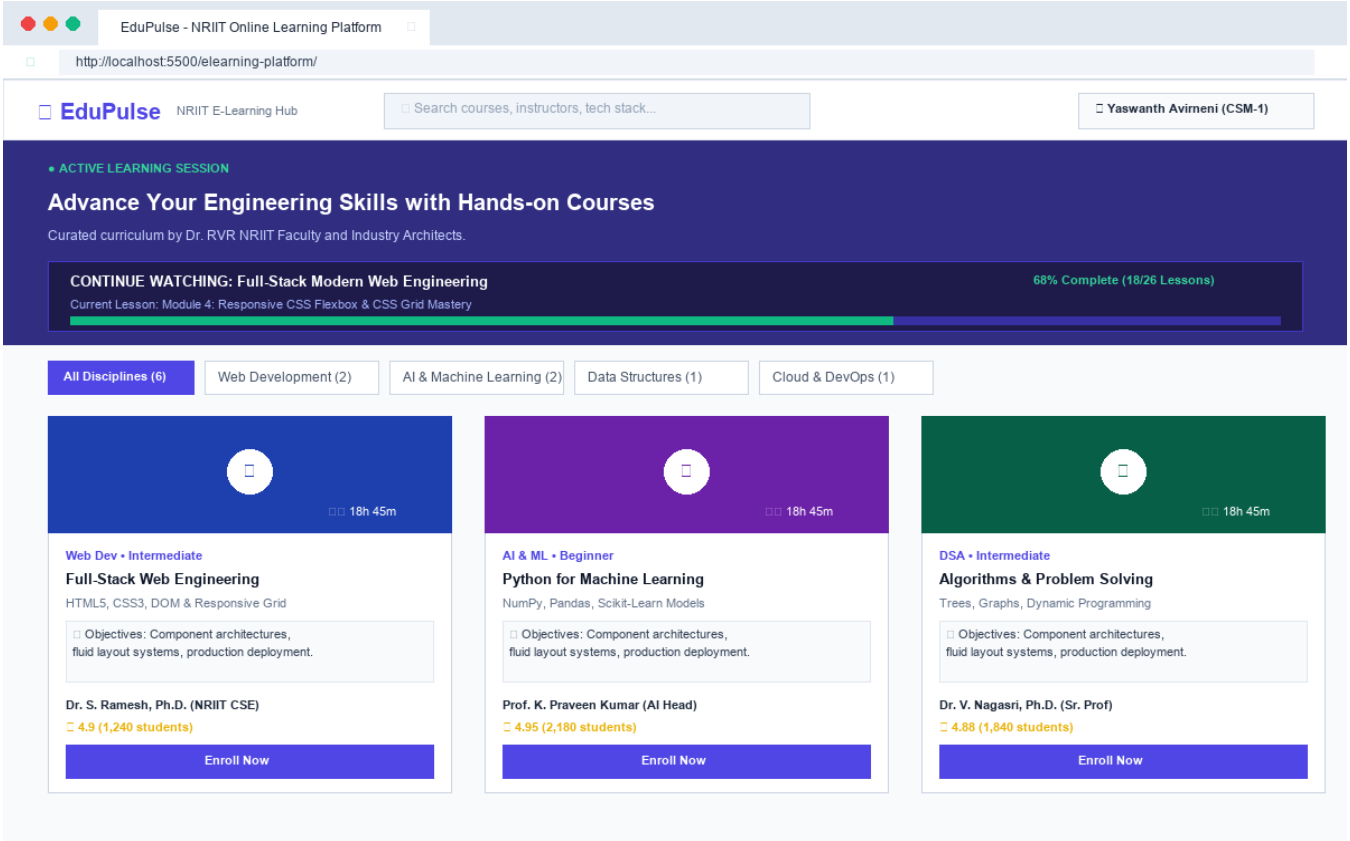
• **Video Lesson Cards:** Rich thumbnail cards with gradient backdrops, play trigger overlays, duration badges (e.g. 18h 45m), and tech-stack chips.

• **Structured Learning Objectives:** Dedicated learning outcomes breakdown for each course (NumPy, Scikit-Learn, PyTorch, Docker, CSS Grid).

• **Pure CSS Modal Interactivity:** Implemented enrollment syllabus popups and enrollment confirmation dialogs using pure CSS :target pseudo-class selectors with zero JavaScript dependency.

• **Instructor Directory & Star Ratings:** Faculty profiles with verified designations, student enrollments, and academic ratings.

Interface Render Screenshot (Online Learning Platform):



3. Case Study 11: Hospital and Healthcare Services Website

Directory: healthcare-portal/ | Files: index.html, style.css

Problem Definition & Scope: Create a tertiary hospital web portal presenting clinical departments, specialist physician directories, diagnostic facilities, visiting hours regulations, 24/7 trauma emergency contact details, and an accessible appointment request form.

Key Technical Capabilities Implemented:

• **24/7 Emergency Trauma Bar:** Urgent red alert header with pulsating status dot and emergency hotline (9640334422 / 1800-NRIIT-CARE).

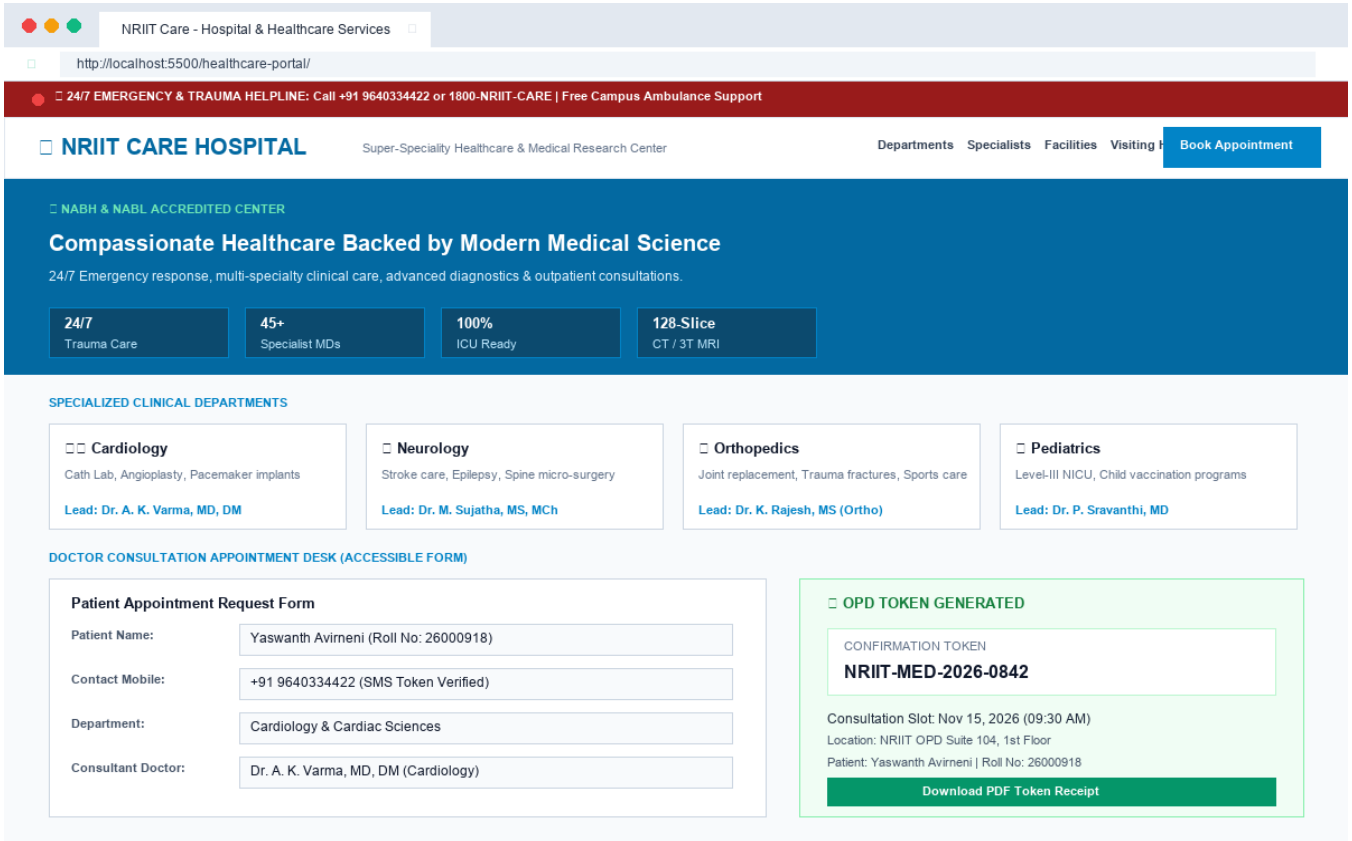
• **Clinical Departments Grid:** Comprehensive medical department cards (Cardiology, Neurology, Orthopedics, Pediatrics, Pulmonology, Intensive Care).

• **Doctor Specialist Directory:** Detailed consultant physician profiles with credentials, OPD suite numbers, consultation schedules, and availability tags.

• **Inpatient Visiting Hours Matrix:** Structured table displaying ward-specific visiting windows, attendant limits, and visitor pass regulations.

• **Accessible Appointment Request Form:** Semantic HTML form with accessible <label> tags, pattern-matched phone number inputs, required selects, date pickers, priority radios, and pure CSS confirmation receipt.

Interface Render Screenshot (Hospital & Healthcare Services):



4. Academic Evaluation Rubric & Faculty Assessment Sheet

Web Development Activity-Based Learning Evaluation (Total: 10 Marks)

Evaluation Parameter	Max Marks	Implementation Standard Achieved	Awarded
Semantic HTML5 Structure	2.5	Extensive use of semantic tags (<header>, <aside>, <main>, <section>, <article>, <table>, <form>).	___ / 2.5
CSS3 Styling, Flexbox & Grid	2.5	Modern CSS custom properties, responsive multi-column layouts, smooth card hover transitions, and progress bar animations.	___ / 2.5
Responsive UI & Accessibility	2.5	Fluid media queries across mobile, tablet, and desktop viewports; accessible form controls and clear visual hierarchy.	___ / 2.5
Functional Completeness & Case Coverage	2.5	Exact alignment with Case Studies 4, 3, and 11 from the Dr. RVR NRIIT curriculum sheet, documentation, and repository.	___ / 2.5
Total Score	**10.0**	**Comprehensive Web Development Activity Submission**	**___ / 10**

Student Signature:

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Section: CSM - 1
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Evaluator / Faculty Signature:

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